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Poll Everywhere

Consider a curve given by the parametrization  $x = f(t)$  and  $y = g(t)$ . If  $\frac{dy}{dt} \big|_{t=a} \neq 0$  and  $\frac{dx}{dt} \big|_{t=a} = 0$ , what can we tell about the tangent line to the curve at the point given by  $t = a$ ?



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There is no tangent line at that point.

The tangent line is a horizontal line.

The tangent line is a vertical line.

The slope of the tangent line is exactly equal to  $\frac{dy}{dt} \big|_{t=a}$ .

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Poll Everywhere

We want to find the area under  $y = x^3$  over  $[0, 1]$

. Choose the correct expression for the area using the parametrization  $x = t^2$ ,  $y = t^6$ ,  $0 \leq t \leq 1$ .



0



0

$$\int_0^1 2t^6 dt$$

$$\int_0^1 2t^7 dt$$

$$\int_0^1 6t^6 dt$$

$$\int_0^1 6t^7 dt$$

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Poll Everywhere

Choose the correct formula for the length of a parametrized curve for  $a \leq t \leq b$ .



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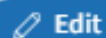
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$$\int_a^b \sqrt{1 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\int_a^b \sqrt{1 + (dy/dx)^2} dt$$

$$\int_a^b \sqrt{\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2}} dt$$

$$\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$



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Poll Everywhere

Which polar coordinate pairs label the same point?



0



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$(2, -\frac{\pi}{3})$  and  $(-2, \frac{\pi}{3})$

$(-2, \frac{\pi}{3})$  and  $(2, -\frac{2\pi}{3})$

$(r, \theta)$  and  $(r, \theta + \pi)$

$(-r, \theta)$  and  $(-r, \theta + \pi)$

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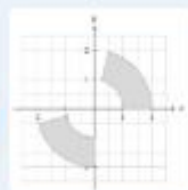
Choose the correct graph for  $0 \leq \theta \leq \pi/2$ ,  $1 \leq |r| \leq 2$

0 0



A

0%



B

0%



C

0%



D

0%





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Poll Everywhere

When converting cartesian equation  $(x - 5)^2 + y^2 = 25$  to polar equation, I write

$$x^2 - 10x + 25 + y^2 = 25 \implies x^2 + y^2 = 10x \implies r^2 = 10r \cos \theta \implies r = \pm \sqrt{10r \cos \theta}$$

. What do you think about this argument?



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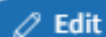
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Sounds good to me.

It is flawed.

It is flawed but I cannot find a reason for that.

It is technically correct but it gives no information on the values of  $r$ .



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